

Naima Khan

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Summary

Data Scientist with experience in applying statistical methods and machine learning to solve complex data challenges. Proven expertise in Python-based analytics and prototype development in both academic and industry settings. Adept at collaborating with cross-functional teams to transform business objectives into actionable insights and structured data problems.

Key Skills

- **Programming:** Python, R, C, C++, Matlab
- **ML/DS:** Keras, PyTorch, Tensorflow, SciPy, Sk-learn, Pandas, Matplotlib, Machine Learning, Data Visualization, Statistical Techniques
- **Data Cloud:** PostgreSQL, SQL, AWS, Google Cloud Platform, Docker, Cloud Development
- **Tools/ Softwares:** Anaconda, Git, Bitbucket, Eclipse, Jupyter Notebook, PyCharm, Tableau, Excel, Weka
- **ML/DL Concepts:** Structured and unstructured data analysis, modeling, Time-series forecasting, Normalization, Regression, Classification, Clustering, Supervised and Unsupervised Learning, Random Forest, Decision Trees, XGBoost, SVM, Bootstrapping, K-means, Feature Engineering, Deep learning, Computer vision, Object segmentation & detection, Anomaly Detection, Meta learning, Descriptive Analytics, Predictive Analytics, Prescriptive Analytics

Work Experience

US FOOD AND DRUG ADMINISTRATION (FDA).

Jan 2024 - Mar 2025

Postdoctoral ORISE Research Fellow

- Engineered a meta-learning based machine learning model to detect gait events (heel strike and toe off) from an imbalanced image dataset, boosting detection accuracy by 40% while applying Python and predictive analytics to transform complex research into actionable metrics.
- Evaluated deep learning models (OpenPose, MediaPipe) for gait metric extraction and processed multi-modal gait signals from wearable sensors and video devices, supporting data-driven decision making and cross-functional collaboration.

FINRA.

May 2022 - Aug 2022

Data Science Intern

- Executed unsupervised anomaly detection on AWS VPC flowlog data and devised clustering-based approaches to infer network flow anomalies, utilizing Python and statistical methods to align technical outcomes with business objectives.
- Developed unit tests for a machine learning pipeline and explored data parallelism to expedite deep learning model training, reinforcing robust model performance and adherence to cloud development best practices.

UNIVERSITY OF MARYLAND BALTIMORE COUNTY.

Jan 2018 - Dec 2023

Graduate Research/ Teaching Assistant

- Formulated a thermal anomaly detection model to pinpoint abnormal thermal variations on building surfaces using longitudinal thermal images, enhancing detection accuracy by 30% with statistical techniques and predictive analytics.
- Implemented an unsupervised temporal clustering model to generate customized thermal profiles and improve cluster consistency in slowly evolving temperature data, providing actionable insights for cross-functional team evaluations.
- Developed an LSTM-based approach to identify thermal condition change events from temperature sensor data, translating qualitative observations into quantitative experimental models.
- Pioneered unsupervised multi-modal learning methodologies for simultaneous analysis of thermal and humidity images, deriving insights that informed data logging requirements and experimental evaluations.
- Facilitated programming courses in Java and C, fostering technical proficiency, effective communication, and collaborative problem-solving.

Education

University of Maryland Baltimore County (UMBC)

2018 - Nov 2023

Doctor of Philosophy, Information Systems

- **GPA:** 3.84/4.0
- **Achievements:** Dissertation Fellowship, University of Maryland Baltimore County
- **Coursework:** Online Exploratory Data Analysis, AWS machine Learning, Analysis of Algorithms, Computer Simulation, Operating Systems, Theory and Implemental Database, Cryptography and Data Security, Data Mining and Machine Learning, Deep Learning, Information Retrieval, Natural Language Processing, Smart Home Health Analytics, Probabilistic Machine Learning, Generative AI, Computational Methods, Advanced Quantitative Methods

Bangladesh University of Engineering and Technology (BUET)

2008 - 2013

Bachelor of Science, Computer Science and Engineering

Certifications

- Introduction to Machine Learning on AWS, Python and Pandas for Data Engineering, Automation Scripts Using Bash

Key Publications

- Anderson, Anthony J., Michael Gonzalez, David Eguren, Naima Khan, Isabella Zuccaroli, Siegfried Hirczy, Valerie Kelly, Brittney Muir, and Kimberly L. Kontson. Deep learning-based stride segmentation with wearable sensors: effects of data quantity, sensor location, and task..medRxiv (2024)
- Anderson, Anthony J., David Eguren, Michael A. Gonzalez, Naima Khan, Sophia Watkinson, Michael Caiola, Siegfried S. Hirczy et al..WearGait-PD: An Open-Access Wearables Dataset for Gait in Parkinson's Disease and Age-Matched Controls..medRxiv (2024): 2024-09
- Khan, Naima, Md Abdullah Al Hafiz Khan, Nirmalya Roy. Unsupervised spatio-temporal anomalous thermal behavior monitoring of inside-built environments..2023 Annual International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT 2023). IEEE, 2023. (Best paper award)
- Khan, Naima, Masud Ahmed, and Nirmalya Roy. Temporal Clustering Based Thermal Condition Monitoring in Building..Sustainable Computing: Informatics and Systems 29 (2021): 100441. (Best paper award)

Awards

- **Best Paper Award:** Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT).
- **Best Paper Award:** International Green and Sustainable Computing Conference (IGSC).
- **Dissertation Fellowship:** University of Maryland Baltimore County.
- **NSF Travel Grant:** Pervasive Computing and Communications (PerCom).
- **NSF Travel Grant:** International Conference on Smart Computing (SMARTCOMP).
- **NSF Travel Grant:** International Green and Sustainable Computing Conference (IGSC).